

Introduction to Data Visualization in Stata

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Workshop Agenda

- Five Principles of visualization
- Histograms
- Bar charts
- Scatterplots
- Saving graphs as images

We will work with the classic Iris dataset, loaded directly from GitHub.

Five Principles of Visualization

Principle 1: Show the Data

- Highlight the data that matter most to your argument or story.
- Avoid overwhelming readers with unnecessary detail that obscures important patterns.
- Use visual emphasis to distinguish key observations from background data.

The goal is not to show less data, but to make the most important data stand out.

Source: Schwabish, *Better Data Visualizations: A Guide for Scholars, Researchers, and Wonks.*

Principle 2: Reduce the Clutter

- Remove unnecessary visual elements that distract from the data.
- Avoid excessive gridlines, tick marks, labels, and decorative effects.
- Use simple colors and avoid unnecessary 3D effects that distort data.

Simplify your visualization so readers can focus on the data and its key patterns.

Source: Schwabish, *Better Data Visualizations: A Guide for Scholars, Researchers, and Wonks*.

Principle 3: Integrate the Graphics and Text

- Integrate text and visual elements to help readers interpret the data.
- Label data directly when possible, rather than relying on separate legends.
- Use informative titles that communicate the graph's main takeaway.
- Add annotations to explain important patterns, outliers, or trends.

Help readers understand both what the data show and why it matters.

Source: Schwabish, *Better Data Visualizations: A Guide for Scholars, Researchers, and Wonks*.

Principle 4: Avoid the 'Spaghetti Chart'

- Avoid cramming too much information into a single visualization.
- Use small multiples to split complex data into smaller, comparable charts.
- Keep scales, layouts, and colors consistent across panels.
- Arrange charts logically and prioritize overall patterns over excessive detail.

When one graph becomes too crowded, use multiple, simpler visuals to reveal patterns.

Source: Schwabish, *Better Data Visualizations: A Guide for Scholars, Researchers, and Wonks*.

Principle 5: Start with Gray

- Begin with gray data elements before adding color.
- Use color strategically to highlight important observations or patterns.
- Emphasize key data with selective color, labels, or line thickness.
- Keep background information visually subdued so the main findings stand out.

Use visual emphasis purposefully to direct readers' attention to what matters most.

Source: Schwabish, *Better Data Visualizations: A Guide for Scholars, Researchers, and Wonks*.

Histograms

When to use histograms:

When you want to visualize the **distribution** of a single continuous variable by grouping values into bins.

- Where are values concentrated, and how spread out are they?
- Help spot skewness, gaps, and outliers.
- Compare distributions across groups (e.g., by species).

Stata command:

```
histogram myvariable
```

Bar Charts

When to use bar charts:

When you want to visualize the comparison of a **summary statistic** (mean, count, median, min, max, SD) across categories.

- Compare groups at a glance.
- Works with any statistic Stata can compute.

Stata command:

```
graph bar (summary statistic) myvariable, over(groupvariable)
```

Scatterplots

When to use scatterplots:

When you want to visualize the **relationship between two continuous variables**, using one point per observation.

- Detect trends, clusters, and outliers.
- Judge the direction and strength of a relationship.

Stata command:

```
twoway scatter myvariable1 myvariable2
```

Saving Graphs as Images

What it's used for:

Exporting finished graphs to PNG so they can go into papers, slides, and reports.

- Files save to your working directory
- Set a custom `width()` for higher resolution

Stata command:

```
graph export "graph_name.png", as(png) width(2000) replace
```

Additional Resource

Graphs Everyone Should Know and How to Create Them in Stata

A practical resource by Franz Buscha for exploring different visualization forms and how to create them in Stata.

Recommended reading for exploring additional graph types and techniques.

